

prettytex
write prettier L^AT_EX

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1 General conventions and ideas

This document is, in theory, only a nice-to-have as all relevant information *should* be written in the corresponding style files. However.

1.1 Introduction

This collection of macros and classes was started during bachelor programmes (Physics and Information and Computer Engineering) for homework assignments and lab-reports. Based on the commit history, we started in 2020, but it is likely that some forms of the collection were individually started earlier.

The goal of *prettytex* is to make writing pretty (thus the name) L^AT_EX documents easy. This is achieved by outsourcing things like package loading, page layout, colours, macros, operators and such to macro-collections grouped into different packages, which are:

- | | | |
|------------------|------------------|-----------------|
| • <i>base</i> | • scriptum | • math-sigtrans |
| • <i>math</i> | • invoice | • math-theorems |
| • gfx | • code | • physics |
| • mathematicians | • contract | • pseudo |
| • shortcuts | • neturalThesis | • thesis |
| • theorems | • gfx-components | • todo |
| • boxes | • gfx-circuits | |

Packages highlighted in **red** are bound to be removed from the main branch, packages marked **orange** are put on the back-burner in the refactor, while a **purple** highlight means, that the package is going to be refactored.

1.2 Package options

We provide some elementary package options, which are handled via *xkeyval*. We have some key-value options, like lang in *base*, but also quite a few boolean-flags, like vargreek in *math*. In order to pass a key-value option, load the package as such:

```
\usepackage[<option>=<value>]{prettytex/<package>}
```

To set a boolean-flag to it's non-default value (every boolean flag has a default which is either **true** or **false**), simply pass the option name as such:

```
\usepackage[<booleanFlag>]{prettytex/<package>}
```

1.3 *prettytex* macros

Each package typically provides a few so called *prettytex-macros*. These are used to keep macros, operators and related things modular to a certain degree. This concept is heavily used in *math*. For example, currently, the `\trace` operator results in trace. However, if you wish to alternatively use, say, `tr`, instead of redefining the `\trace` operator, you can redefine `\prettytexMathTrace` to `tr` and be done with it. *prettytex-macros* follow a very simple syntax:

```
\prettytex<Package><Macro> ,
```

so all *prettytex-macros* in *math* are of the form `\prettytexMath<Macro>`. The exception here is *base*, where it simply is `\prettytex<Macro>`. *prettytex* macros are always listed in their corresponding package description. To keep everything legible, we will mostly omit the `\prettytex<Package>` when talking about *prettytex-macros*, thus we write *prettytex-macros* in colour, so **Trace** is actually `\prettytexMathTrace`.

1.4 Convention on argument naming in the documentation

Many macros use arguments, either mandatory or optional. Most of these can be any valid tex-expression (depending on the mode either text or math), however to highlight their intended use, we name arguments in a certain manner. For example, the argument of a formatting macro will be labelled `text`, while for general mathematical expressions we use `expression`. Optional arguments are highlighted in two ways: either with (o) or (O) (this loosely follows the convention of *xparse*), where (o) means the argument is optional, and (O) optional, but a default value is provided.

argument type	description
<code>text</code>	text argument
<code>colour</code>	<i>xcolor</i> colour identifier
<code><number type></code>	can be <code>integer</code> or <code>float</code>
<code><signed> <number type></code>	the sign can be one of <code>positive</code> (> 0), <code>negative</code> (< 0), <code>nonpositive</code> (≤ 0) or <code>nonnegative</code> (≥ 0)
<code><number type> in <set></code>	<code><set></code> can be any set, typically used for intervals like <code>[1,26]</code>
<code>expression</code>	any valid math-mode expression
<code>array</code>	valid array-contents (anything that fits into <code>bmatrix</code> and <code>relatives</code> , as well as a plain array in math mode etc)
<code>reference</code>	valid reference identifier for labels

Table 1: Argument types

Some macros allow multiple argument types, these are separated by an `or` in the argument list, see for example `cw` in table 7. If a macro uses multiple arguments, they are listed in the expected order and separated by commas.

2 base

base is the entry-point of *prettytex*. It is required by all other packages. It is used to set global options like language, page layout and such and loads elementary packages.

2.1 Package options

option	description	type
lang	the language to use throughout the document (mainly used for babel), expect valid babel language	string
singleLineCheck	the package <i>caption</i> provides the option <code>singlelinecheck</code> which allows to toggle whether or not captions of floats are centered (indented) or not via a boolean flag. false disables centered captions, default is true	boolean
colouredLinks	whether or not to use colouredLinks for reference, hyperrefs etc. default is true . If false , links will instead be highlightet by a coloured underline	boolean
parSkip	the <i>parskip</i> package introduces some small amount of vertical space between paragraphs instead of indenting the new paragraph. If true , the pacakge is loaded. Default is false	boolean
beamer	the <i>beamer</i> package clashes heavily with external page layouts, this if this option is passed, our page layout is disabled. Default is false	boolean

Table 2: Package options for *base*

2.2 Loaded packages

package	description	options
<i>fontenc</i>	T1 font encoding ensures that non-ascii characters like the german umlaute (ö,ä etc) are actual glyphs instead of accented letters. This also improves copy-paste behavior	T1
<i>inputenc</i>	allows for utf8 characters to bes used in source files (e.g. ä.ö. etc)	utf8
<i>lmodern</i>	use latin modern font instead of computer modern	
<i>float</i>	allows for better control of float placement	
<i>geometry</i>	used for page layout (mostly for margins)	
<i>xcolor</i>		table
<i>cellspace</i>		column=C
<i>hyperref</i>	place clickable links in the text (if enabled via options, links are coloured instead of underlined)	hypertextnames=false
<i>translations</i>	allows for automatic translations of text (e.g. theorem names)	
<i>contour</i>	give text an outline of any colour and width	outline
<i>xspace</i>		
<i>subcaption</i>		
<i>graphicx</i>		
<i>parskip</i>	only loaded if enabled via option, see 2 for explanation	
<i>enumitem</i>	more control over enums, specifically their labelling, only loaded if beamer is false	inline

Table 3: Packages loaded by *base*

2.3 Document geometry

The options passed to *geometry* are

- a4paper,
- left=2cm,
- right=2cm,
- top=1.25cm,
- bottom=1.25cm,
- includeheadfoot

2.4 Colours

prettytex provides a few colours:

color-name	defined as	R	G	B	hex-code
purple	RGB	180	2	212	#b402d4
green	RGB	0	153	51	#009933
red	RGB	204	0	0	#cc0000
blue	RGB	0	102	255	#0066ff
orange	RGB	227	102	0	#e36600
darkred	HTML				#e05263
darkpurple	HTML				#5a016a
pastelpurple	HTML				#a580d1
pastelpink (pastelpink)	HTML				#f8c8dc
lightgrey (lightgrey)	HTML				#dddddd
darkpink	HTML				#e6acac
themecolor	HTML				#00a8ff
themecolor2	HTML				#659157
themecolor3	HTML				#3d518c
themecolor4	HTML				#f25f5c
lightgreybg (lightgreybg)	rgb	0.97	0.97	0.97	
headrulecolor (headrulecolor)	rgb	0.8	0.8	0.8	
sigmared	HTML				#af0f1e
sigmayellow	HTML				#ffd500
discord	HTML				#5865f2

Table 4: Colours defined by *prettytex*

2.5 Formatting

We also provide some formatting macros:

macro	expands to
HighlightColour	pastelpurple
InfoColour	themecolor
WarnColour	orange
ErrorColour	red

Table 5: *prettytex*-macros

macro	arguments	description	example
\ebf	text	emphasised and bold-faced text	<i>example</i>
\ibf	text	italicised and bold-faced text	<i>example</i>
\ibf	text	underlined and bold-faced text	<u>example</u>
\eubf	text	emphasised, underlined and bold-faced text	<u>example</u>
\iubf	text	italicised, underlined and bold-faced text	<u>example</u>
\col	colour, text	write text in colour	example
\REFL	text	highlight reference that does not yet exist	example
\highlight	text	highlight text	example
\info	text	info text	example
\warn	text	warn text	example
\error	text	error text	example

Table 6: Formatting macros

macro	arguments	description	example/output
<code>\cw</code>	text or expression	give the passed argument a white outline with width .06em	example
<code>\th</code>		write th as superscript (for dates)	4 th
<code>\prettytex</code>		write prettytex in pacakge format	<i>prettytex</i>

Table 7: Miscellaneous macros

2.6 Miscellaneous

macro	arguments	description	example/output
<code>\tightlist</code>		remove vertical spacing from all lists	
<code>\rom</code>	positive integer	convert input to lowercase roman numeral	xlii
<code>\Rom</code>	positive integer	convert input to uppercase roman numeral	LXIX
<code>\alp</code>	integer in [1,26]	convert input to lowercase alph-value	x
<code>\Alp</code>	integer in [1,26]	convert input to uppercase alph-value	P
<code>\listfigures</code>		this is a verbatim copy of the built-in macro <code>\listfigures</code> but allows the list to be included as a subsection	
<code>\listtables</code>		this is a verbatim copy of the built-in macro <code>\listtables</code> but allows the list to be included as a subsection	
<code>\autoheadings</code>		set header and footer to sensible default based on document metadata	

Table 8: Miscellaneous macros from *base*

3 *math*

math is the package for all¹ things mathematics.

3.1 Package options

option	description	type
vargreek	if true <code>\rho</code> and <code>\epsilon</code> are redefined to their variants <code>\varrho</code> and <code>\varepsilon</code>	boolean

Table 9: Package options for *math*

3.2 Loaded packages

package	description	options
<i>amsmath</i> <i>amssymb</i> <i>esint</i> <i>bm</i> <i>upgreek</i> <i>nicematrix</i>	provides non-italicised greek letter	

Table 10: Packages loaded by *math*

3.3 Fonts and alphabets

We declare a new math-alphabet (`\mathscall`) for the lowercase o for Landau-symbols. As a shortcut, we already provide the `\sco` macro (which is an abbreviation of small calligraphic o) and results in *o*.

3.4 Formatting

Sometimes we need to write text in equations and keep a nice format. Thus some formatting macros were created:

macro	arguments	description	example/output
<code>\qtext</code>	text	writes <code>text</code> between two <code>\quad</code> , use in display-mode to separate expressions with natural language same as <code>qtext</code> but with <code>\quad</code> instead of <code>\quad</code>	<i>example</i>
<code>\qtext</code>	text		example
<code>\bar</code>	expression	put a bar over the expression	$\bar{z}$
<code>\ubar</code>	expression	underline the expression	$\underline{z}$
<code>\oset</code>	expression, expression	types the first expression over the second, this is just a shortcut for <code>\overset</code>	$\overset{\rho}{=}$
<code>\uset</code>	expression, expression	same as <code>oset</code> , but now underset, of which this is a shortcut	$\underset{\rho}{=}$
<code>\obra</code>	expression	put a brace over the expression (allows for proper superscript placement)	$\overbrace{e^{i\pi}}$
<code>\ubra</code>	expression	put a brace under the expression (allows for proper subscript placement)	$\underbrace{\sin^2(x) + \cos^2(x)}$
<code>\parenth</code>	expression	put automatically scaling parenthesis around the expression	$\lambda \left(\int_0^1 f(x) dx \right)$
<code>\eqv</code>		equivalence arrow from logic	$\Leftrightarrow$

Table 11: Formatting macros for mathematics

¹well, almost all

3.5 Sets

macro	expands to	result
Complement	<code>\mathsf{c}</code>	c

Table 12: *prettytex-math* macros for sets

macro	arguments	description	example/output
<code>\set</code>	expression	shortcut for set-builder notation, places automatically scaling braces around the expression	$\{q \in \mathbb{Q} \mid q^2 < 2\}$
<code>\sm</code>		alias for <code>\setminus</code>	
<code>\compl</code>		complement of a set, write this after the set whose complement you need	A^c
<code>\eset</code>		alias for <code>\emptyset</code>	$\emptyset$
<code>\sset</code>		alias for <code>\subseteq</code>	$\subseteq$
<code>\psset</code>		alias for <code>\subset</code>	$\subset$
<code>\Sset</code>		alias for <code>\supseteq</code>	$\supseteq$
<code>\pSset</code>		alias for <code>\supset</code>	$\supset$
<code>\inv</code>		places -1 in the superscript of the previous expression (use like <code>\compl</code>)	f^{-1}

Table 13: Set macros from *math*

3.6 Numbers and relatives

macro	expands to	result
ComplexArgument	<code>Arg</code>	Arg
Real	<code>\mathfrak{Re}</code>	$\Re$
Imag	<code>\mathfrak{Im}</code>	$\Im$
AbsoluteValuePlaceholder	<code>\cdot</code>	$\cdot$

Table 14: *prettytex-math* macros for numbers

macro	arguments	description	example/output
<code>\N</code>		the natural numbers	$\mathbb{N}$
<code>\Z</code>		the integers	$\mathbb{Z}$
<code>\Q</code>		the rationals	$\mathbb{Q}$
<code>\R</code>		the reals	$\mathbb{R}$
<code>\C</code>		the complex numbers	$\mathbb{C}$
<code>\I</code>		the irrational numbers	$\mathbb{I}$
<code>\D</code>		the open unit disk	$\mathbb{D}$
<code>\T</code>		the transcendental numbers	$\mathbb{T}$
<code>\S</code>		the unit sphere	$\mathbb{S}$
<code>\H</code>		the upper half plane	$\mathbb{H}$
<code>\P</code>		the primes or a probability measure	$\mathbb{P}$
<code>\K</code>		general field (german spelling)	$\mathbb{K}$
<code>\F</code>		general field	$\mathbb{F}$
<code>\i</code>		imaginary unit	i
<code>\j</code>		jmaginary unit (electrical engineering)	j
<code>\e</code>		Euler's number	e
<code>\floorl</code>	expression	floor function of expression with scaling delimiters	$\left\lfloor \int f(x) \, dx \right\rfloor$
<code>\floor</code>	expression	floor function of expression	$\lfloor \pi \rfloor$
<code>\ceil</code>	expression	ceiling function of expression	$\lceil e \rceil$
<code>\ceill</code>	expression	ceiling function of expression with scaling delimiters	$\left\lceil \int f(x) \, dx \right\rceil$
<code>\abs</code>	expression	absolute value of expression	$ -2 $
<code>\absc</code>		absolute value of AbsoluteValuePlaceholder , used for definition	$ \cdot : \mathbb{C} \rightarrow \mathbb{R}_0^+$
<code>\labs</code>	expression	absolute value of expression with scaling delimiters	$\left \int f(x) \, dx \right $

Table 15: Number macros from *math*

3.7 Topology

macro	expands to	result
Interior	<code>int</code>	<code>int</code>
Closure	<code>cls</code>	<code>cls</code>
Diameter	<code>diam</code>	<code>diam</code>
Exterior	<code>ext</code>	<code>ext</code>
Support	<code>supp</code>	<code>supp</code>
Frontier	<code>Fr</code>	<code>Fr</code>

Table 16: *prettytex-math* macros for topology

operator	description	example/output
<code>\interior</code>	interior of a set	<code>int S, int(S)</code>
<code>\cls</code>	closure of a set	<code>cls S, cls(S)</code>
<code>\diam</code>	diameter of a set	<code>diam S, diam(S)</code>
<code>\supp</code>	support of a function	<code>supp f, supp(f)</code>
<code>\frontier</code>	frontier of a set	<code>Fr S, Fr(S)</code>

Table 17: *prettytex-math* operators for topology

macro	arguments	description	example/output
<code>\intr</code>	expression	interior of expression (accents with a small circle)	$\mathring{S}$
<code>\inter</code>	expression	alias to <code>\intr</code>	$\mathring{S}$
<code>\fc</code>		collection of convergent sequences	$\mathfrak{c}$
<code>\fU</code>		collection of neighbourhoods	$\mathfrak{U}$
<code>\fB</code>		base of topology	$\mathfrak{B}$
<code>\fS</code>		subbase of topology	$\mathfrak{S}$

Table 18: Macros from *math* for topology

3.8 Measure Theory

macro	expands to	result
Volume	<code>vol</code>	vol

Table 19: *prettytex-math* macros for measure theory

operator	description	example/output
<code>\leb</code>	Lebesgue measure	$\lambda, \lambda M, \lambda(M)$
<code>\borel</code>	Borel measure	$\beta, \beta M, \beta(M)$
<code>\me</code>	general measure	$\mu, \mu M, \mu(M)$
<code>\vol</code>	volume of a set	$\text{vol}, \text{vol } M, \text{vol}(M)$

Table 20: *prettytex-math* operators for measure theory

3.9 Derivatives

macro	expands to	result
Hessian	<code>H</code>	H
Jacobian	<code>J</code>	J
Gradient	<code>grad</code>	grad
Laplacian	<code>\triangle</code>	$\triangle$

Table 21: *prettytex-math* macros for derivatives

operator	description	example/output		
		alone	w/o parenthesis	w parenthesis
<code>\hess</code>	hessian of a differentiable function	H	$H f$	$H(f)$
<code>\jac</code>	jacobian of a differentiable function	J	$J f$	$J(f)$
<code>\grad</code>	gradient of a differentiable function	grad	$\text{grad } f$	$\text{grad}(f)$
<code>\laplace</code>	laplacian of a differentiable function	$\triangle$	$\triangle f$	$\triangle(f)$
<code>\argmin</code>	minimiser of function or set of minima	$\arg \min$	$\arg \min f$	$\arg \min(f)$
<code>\argmax</code>	maximiser of function or set of maxima	$\arg \max$	$\arg \max f$	$\arg \max(f)$
<code>\minimise</code>	used for stating minimisation problems	minimise	$\text{minimise } f$	$\text{minimise}(f)$

Table 22: *prettytex-math* operators for derivatives

macro	arguments	description	example/output
<code>\FD</code>		Fréchet derivative	d
<code>\dd</code>		general derivative symbol	d
<code>\idd</code>		used for differential in integrals (adds space after function)	d
<code>\ddx</code>		shortcut for <code>\dd x</code>	dx
<code>\ddy</code>		shortcut for <code>\dd y</code>	dy
<code>\ddz</code>		shortcut for <code>\dd z</code>	dz
<code>\ddu</code>		shortcut for <code>\dd u</code>	du
<code>\ddv</code>		shortcut for <code>\dd v</code>	dv
<code>\ddr</code>		shortcut for <code>\dd r</code>	dr
<code>\dds</code>		shortcut for <code>\dd s</code>	ds
<code>\ddt</code>		shortcut for <code>\dd t</code>	dt
<code>\derx</code>	expression	derivative with respect to x , Leibniz notation	$\frac{df}{dx}$
<code>\nderx</code>	expression	n -th derivative with respect to x , Leibniz notation	$\frac{d^n f}{dx^n}$
<code>\kderx</code>	expression	k -th derivative with respect to x , Leibniz notation	$\frac{d^k f}{dx^k}$
<code>\nder</code>	expression	n -th derivative, Newton notation	$f^{(n)}$
<code>\kder</code>	expression	k -th derivative, Newton notation	$f^{(k)}$

Table 23: Macros from *math* for derivatives

3.10 Sequences and Limits

macro	arguments	description	example/output
<code>\seqn</code>	expression	sequence indexed with n in $\mathbb{N}$	$(x_n)_{n \in \mathbb{N}}$
<code>\seqnz</code>	expression	sequence indexed with n in $\mathbb{N}_0$	$(x_n)_{n \in \mathbb{N}_0}$
<code>\seqk</code>	expression	sequence indexed with k in $\mathbb{N}$	$(x_k)_{k \in \mathbb{N}}$
<code>\seqkz</code>	expression	sequence indexed with k in $\mathbb{N}_0$	$(x_k)_{k \in \mathbb{N}_0}$
<code>\iseq</code>	expression	general family indexed with i and index set I	$(x_i)_{i \in I}$
<code>\jseq</code>	expression	general family indexed with j and index set J	$(x_j)_{j \in J}$
<code>\net</code>	expression	net (map from a directed set into any set)	$(x_\lambda)_{\lambda \in \Lambda}$
<code>\Kappa</code>		uppercase kappa	$\mathbf{K}$
<code>\subnet</code>	expression	subnet of a net	$(x_{\lambda_\kappa})_{\kappa \in \mathbf{K}}$
<code>\limh</code>		limit as h goes to 0	$\lim_{h \rightarrow 0}$
<code>\limt</code>		limit as t goes to 0	$\lim_{t \rightarrow 0}$
<code>\limx</code>		limit as x goes to x_0	$\lim_{x \rightarrow x_0}$
<code>\limz</code>		limit as z goes to z_0	$\lim_{z \rightarrow z_0}$
<code>\limn</code>		limit as n goes to ∞	$\lim_{n \rightarrow \infty}$
<code>\limk</code>		limit as k goes to ∞	$\lim_{k \rightarrow \infty}$
<code>\liminf n</code>		limit inferior as n goes to ∞	$\liminf_{n \rightarrow \infty}$
<code>\limsup n</code>		limit superior as n goes to ∞	$\limsup_{n \rightarrow \infty}$
<code>\liminf k</code>		limit inferior as k goes to ∞	$\liminf_{k \rightarrow \infty}$
<code>\limsup k</code>		limit superior as k goes to ∞	$\limsup_{k \rightarrow \infty}$
<code>\infn</code>		infimum of family indexed by n in $\mathbb{N}$	$\inf_{n \in \mathbb{N}}$
<code>\infk</code>		infimum of family indexed by k in $\mathbb{N}$	$\inf_{k \in \mathbb{N}}$
<code>\supn</code>		supremum of family indexed by n in $\mathbb{N}$	$\sup_{n \in \mathbb{N}}$
<code>\supk</code>		supremum of family indexed by k in $\mathbb{N}$	$\sup_{k \in \mathbb{N}}$

Table 24: Macros from *math* for limits and sequences

3.11 Algebra

macro	expands to	result
<code>\Homset</code>	<code>Hom</code>	Hom
<code>\Endomorphisms</code>	<code>End</code>	End
<code>\Automorphisms</code>	<code>Aut</code>	Aut
<code>\Range</code>	<code>ran</code>	ran
<code>\Domain</code>	<code>dom</code>	dom
<code>\Codomain</code>	<code>co\:\text{dom}</code>	co dom
<code>\Projection</code>	<code>proj</code>	proj
<code>\Cokernel</code>	<code>coker</code>	coker
<code>\PermutationParity</code>	<code>sign</code>	sign
<code>\Sign</code>	<code>sgn</code>	sgn
<code>\Image</code>	<code>im</code>	im
<code>\Kernel</code>	<code>ker</code>	ker
<code>\Identity</code>	<code>id</code>	id
<code>\Degree</code>	<code>deg</code>	deg

Table 25: *prettytex-math* macros for algebra

macro	arguments	description	example/output
<code>\epimorph</code>		arrow for epimorphism	$\twoheadrightarrow$
<code>\monomorph</code>		arrow for monomorphism	$\rightarrowtail$
<code>\incl</code>		arrow for inclusion	$\hookrightarrow$
<code>\surj</code>		arrow for surjection (alias to <code>\epimorphism</code>)	$\twoheadrightarrow$
<code>\unity</code>		group of units of a structure (write after symbol)	$R^\times$

Table 26: Macros from *math* for algebra

operator	description	example/output		
		alone	w/o parenthesis	w parenthesis
<code>\Hom</code>	homomorphisms between objects	Hom		$\text{Hom}(R, S)$
<code>\End</code>	endomorphisms of an object	End	$\text{End } R$	$\text{End}(R)$
<code>\Aut</code>	automorphisms of an object	Aut	$\text{Aut } R$	$\text{Aut}(R)$
<code>\range</code>	range of an operator (i.e. the image)	ran	$\text{ran } T$	$\text{ran}(T)$
<code>\dom</code>	domain of a morphism	dom	$\text{dom } f$	$\text{dom}(f)$
<code>\codom</code>	codomain of a morphism	co dom	$\text{co dom } f$	$\text{co dom}(f)$
<code>\coker</code>	cokernel of a morphism	coker	$\text{coker } f$	$\text{coker}(f)$
<code>\im</code>	image of a morphism	im	$\text{im } f$	$\text{im}(f)$
<code>\ker</code>	kernel of a morphism	ker	$\text{ker } f$	$\text{ker}(f)$
<code>\id</code>	identity map	id		
<code>\proj</code>	general projection	proj		
<code>\sign</code>	signature (parity) of a permutation	sign	$\text{sign } \pi$	$\text{sign}(\pi)$
<code>\sgn</code>	sign of an expression	sgn	$\text{sgn } x$	$\text{sgn}(x)$
<code>\deg</code>	degree of an object	deg	$\text{deg } p$	$\text{deg}(p)$

Table 27: *prettytex-math* operators for algebra

3.12 Category theory

macro	arguments	description	example/output
<code>\CATEGORY</code>	expression	label of a category	A
<code>\CAT</code>		generic category	C
<code>\Cat</code>		alias for <code>\CAT</code>	C
<code>\CATD</code>		generic category (variation)	D
<code>\CatD</code>		alias for <code>\CATD</code>	D
<code>\LEFTMOD</code>	expression	category of left-modules	$S - \mathbf{Mod}$
<code>\LEFTMODR</code>		shortcut for <code>\LEFTMOD{R}</code>	$R - \mathbf{Mod}$
<code>\FINLEFTMOD</code>	expression	category of finitely generated left-modules	$S - \mathbf{mod}$
<code>\FINLEFTMODR</code>		shortcut for <code>\FINLEFTMOD{R}</code>	$R - \mathbf{mod}$
<code>\RIGHTMOD</code>	expression	category of right-modules	$\mathbf{Mod} - S$
<code>\RIGHTMODR</code>		shortcut for <code>\RIGHTMOD{R}</code>	$\mathbf{Mod} - R$
<code>\FINRIGHTMOD</code>	expression	category of finitely generated right-modules	$\mathbf{mod} - S$
<code>\FINRIGHTMODR</code>		shortcut for <code>\FINRIGHTMOD{R}</code>	$\mathbf{mod} - R$
<code>\VECT</code>	expression	category of vectorspaces	$\mathbf{Vect}_F$
<code>\VECTK</code>		shortcut for <code>\VECT{k}</code>	$\mathbf{Vect}_k$
<code>\FINVECT</code>	expression	category of finite dimensional vectorspaces	$\mathbf{vect}_F$
<code>\FINVECTK</code>		shortcut for <code>\FINVECT{k}</code>	$\mathbf{vect}_k$
<code>\CRING</code>		category of commutative rings	CRing
<code>\SET</code>		category of sets	Set
<code>\GRP</code>		category of groups	Grp
<code>\RING</code>		category of rings	Ring
<code>\FIELD</code>		category of fields	Field
<code>\TOP</code>		category of topological spaces	Top
<code>\TOPB</code>		category of based topological spaces	Top_*
<code>\AB</code>		category of abelian groups	Ab
<code>\Homcat</code>		morphisms between two objects in C	$\mathrm{Hom}_{\mathbf{C}}(A, B)$
<code>\UCAT</code>		generic category (variation)	U
<code>\CATJ</code>		generic category (variation)	J
<code>\CATS</code>		generic category (variation)	S

Table 28: *prettytex-math* macros for category theory

3.13 Linear Algebra

macro	expands to	result
Transpose	t	t
HermitianTranspose	H	H
Trace	trace	trace
Rank	rank	rank
RowRank	rank	rank
ColumnRank	rank	rank
DotProductPlaceholder	$\cdot$	.
NormPlaceholder	$\cdot$	.
Dimension	dim	dim
LinearSpan	span	span
GeneralLinearGroup	GL	GL
SpecialLinearGroup	SL	SL
SpecialUnitaryGroup	SU	SU
SpecialOrthogonalGroup	SO	SO
LpDefaultExponent	p	p
Diagonal	diag	diag
Eigen	Eig	Eig
Gram	Gram	Gram
Spectrum	spec	spec
Annihilator	Ann	Ann
AdjugateMatrix	Adj	Adj
CofactorMatrix	Cof	Cof

Table 29: *prettytex-math* macros for linear algebra

macro	arguments	description	example/output
$\backslash\mathrm{tran}$		transpose of a matrix (write after expression)	A^t
$\backslash\mathrm{itrans}$		inverse of a matrix (write after expression)	A^{-t}
$\backslash\mathrm{htra}$		hermitian transpose of a matrix (write after expression)	A^H
$\backslash\mathrm{vec}$	expression	element of a vectorspace, but meant for coordinate vectors	$\boldsymbol{v}$
$\backslash\mathrm{hvec}$	expression	vector with hat	$\hat{\boldsymbol{v}}$
$\backslash\mathrm{whvec}$	expression	vector with wide hat	$\widehat{\boldsymbol{v}}$
$\backslash\mathrm{tvec}$	expression	vector with tilde	$\tilde{\boldsymbol{v}}$
$\backslash\mathrm{wtvec}$	expression	vector with wide tilde	$\widetilde{\boldsymbol{v}}$
$\backslash\mathrm{bvec}$	expression	vector with overbar	$\overline{\boldsymbol{v}}$
$\backslash\mathrm{uvec}$	expression	vector with underbar	$\underline{\boldsymbol{v}}$
$\backslash\mathrm{zvec}$		zero vector	$\mathbf{0}$
$\backslash\mathrm{mat}$	expression	matrix	A
$\backslash\mathrm{hmat}$	expression	matrix with hat	$\hat{A}$
$\backslash\mathrm{whmat}$	expression	matrix with wide hat	$\widehat{A}$
$\backslash\mathrm{tmat}$	expression	matrix with tilde	$\tilde{A}$
$\backslash\mathrm{wtmat}$	expression	matrix with wide tilde	$\widetilde{A}$
$\backslash\mathrm{barmat}$	expression	matrix with overbar ($\backslash\mathrm{bmat}$ is reserved)	$\overline{A}$
$\backslash\mathrm{ubarmat}$	expression	matrix with underbar	$\underline{A}$
$\backslash\mathrm{bmat}$	float (o), array	shortcut for writing matrices, optional argument sets arraystretch	$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

<code>\fbmat</code>	array	same as <code>\bmat</code> , but uses <code>NiceMatrix</code> instead	$\begin{bmatrix} 1 & \dots & \dots & 1 \\ & \ddots & & \\ & & \ddots & \\ 1 & \dots & \dots & 1 \end{bmatrix}$
<code>\dotp</code>	expression, expression	dot product of the expressions	$\langle v, w \rangle$
<code>\Ldotp</code>	expression, expression	same as <code>\dotp</code> but with scaling delimiters	$\langle \sum_{i=1}^n v_i e_i, e_j \rangle$
<code>\dotpc</code>		dot-product of <code>DotProductPlaceholder</code> , use for definition	$\langle \cdot, \cdot \rangle: V^2 \rightarrow \mathbb{K}$
<code>\Lpdotp</code>	e (O), e (o), e, e	L^p -dot product, first optional argument sets the exponent of the L^p -space and has default value <code>LpDefaultExponent</code> , while the second optional argument can be used to specify the measure/measurable space	$\langle f, g \rangle_{L^p}, \quad \langle f, g \rangle_{L^2},$ $\langle f, g \rangle_{L^2(\mathbb{R}^2, \lambda^2)}$
<code>\lLpdotp</code>	e (O), e (o), e, e	same as <code>\Lpdotp</code> but with scaling delimiters	
<code>\Lpdotpc</code>	e (O), e (o)	<code>\Lpdotp</code> of <code>DotProductPlaceholder</code> , optional arguments work identically to <code>\Lpdotp</code>	
<code>\norm</code>	expression (o), expression	norm of expression, optional argument can be used to label the underlying space	$\ f\ , \ f\ _X$
<code>\lnorm</code>	expression (o), expression	same as <code>\norm</code> but with scaling delimiters	$\ \sum_{i=1}^n \xi_i e_i\ $
<code>\normc</code>		norm of <code>NormPlaceholder</code> , use for definition	$\ \cdot\ : V \rightarrow \mathbb{R}$
<code>\inorm</code>	expression	infinity-norm of expression	$\ f\ _\infty$
<code>\linorm</code>	expression	same as <code>\inorm</code> but with scaling delimiters	$\ \sum_{i=1}^n \xi_i e_i\ _\infty$
<code>\inormc</code>		infinity norm of <code>NormPlaceholder</code> , used for definition	$\ \cdot\ : V \rightarrow \mathbb{R}$
<code>\pnorm</code>	expression (O), expression	p -norm of expression, optional argument defaults to <code>LpDefaultExponent</code> and is used to set the exponent	$\ x\ _p, \ x\ _2$
<code>\lpnorm</code>	expression (O), expression	same as <code>\pnorm</code> but with scaling delimiters	
<code>\pnormc</code>	expression (O)	p -norm of <code>NormPlaceholder</code> , optional argument sets the exponent and defaults to <code>LpDefaultExponent</code> , use for definition	$\ \cdot\ _p: \mathbb{R}^n \rightarrow \mathbb{R}$
<code>\Lpnorm</code>	e (O), e (o), e	L^p -norm of expression, first optional argument sets the exponent and defaults to <code>LpDefaultExponent</code> , second optional argument can be used to set the measure/measurable space	$\ f\ _{L^p}, \quad \ f\ _{L^2},$ $\ f\ _{L^2(\mathbb{R}^2, \lambda^2)}$
<code>\lLpnorm</code>	e (O), e (o), e	same as <code>\Lpnorm</code> but with scaling delimiters	
<code>\Lpnomrc</code>	e (O), e (o)	L^p -norm of <code>NormPlaceholder</code> , optional arguments work identically to <code>\Lpnorm</code> , use for definition	$\ \cdot\ _{L^p}: L^p \rightarrow \mathbb{R}$

Table 30: *prettytex-math* macros for linear algebra

operator	description	example/output		
		alone	w/o parenthesis	w parenthesis
<code>\lspan</code>	linear span of collection of elements	span		$\text{span}(v_1, \dots, v_n)$

<code>\dim</code>	dimension of object	<code>dim</code>	$\dim V$	$\dim(V)$
<code>\GL</code>	general linear group of a space	<code>GL</code>	$\mathrm{GL} V$	$\mathrm{GL}(V)$
<code>\SL</code>	special linear group of a space	<code>SL</code>	$\mathrm{SL} V$	$\mathrm{SL}(V)$
<code>\SU</code>	special unitary group of a space	<code>SU</code>	$\mathrm{SU} V$	$\mathrm{SU}(V)$
<code>\SO</code>	special orthogonal group of a space	<code>SO</code>	$\mathrm{SO} V$	$\mathrm{SO}(V)$
<code>\diag</code>	diagonal matrix from elements or diagonal of a matrix	<code>diag</code>	$\mathrm{diag} M$	$\mathrm{diag}(v_1, \dots, v_m)$
<code>\Eig</code>	eigendata of an operator	<code>Eig</code>	$\mathrm{Eig} T$	$\mathrm{Eig}(T)$
<code>\Gram</code>	Gram matrix of vectors	<code>Gram</code>		$\mathrm{Gram}(v_1, \dots, v_n)$
<code>\spec</code>	spectrum of an operator	<code>spec</code>	$\mathrm{spec} T, \mathrm{spec}(T)$	
<code>\Ann</code>	annihilator of object	<code>Ann</code>	$\mathrm{Ann} M$	$\mathrm{Ann}(M)$
<code>\Adj</code>	adjugate of a matrix	<code>Adj</code>	$\mathrm{Adj} M$	$\mathrm{Adj}(M)$
<code>\Cof</code>	cofactor matrix	<code>Cof</code>	$\mathrm{Cof} M$	$\mathrm{Cof}(M)$
<code>\trace</code>	trace of an operator	<code>trace</code>	$\mathrm{trace} T$	$\mathrm{trace}(T)$
<code>\Trace</code>	alias of <code>\trace</code>			
<code>\tr</code>	alias of <code>\trace</code>			
<code>\Tr</code>	alias of <code>\trace</code>			
<code>\spur</code>	alias of <code>\trace</code> (german spelling)			
<code>\Spur</code>	alias of <code>\trace</code> (german spelling)			
<code>\spr</code>	alias of <code>\trace</code> (german spelling)			
<code>\Spr</code>	alias of <code>\trace</code> (german spelling)			
<code>\rank</code>	rank of operator	<code>rank</code>	$\mathrm{rank} T$	$\mathrm{rank}(T)$
<code>\Rank</code>	alias of <code>\rank</code>			
<code>\rk</code>	alias of <code>\rank</code>			
<code>\Rk</code>	alias of <code>\rank</code>			
<code>\rang</code>	alias of <code>\rank</code> (german spelling)			
<code>\Rang</code>	alias of <code>\rank</code> (german spelling)			
<code>\rg</code>	alias of <code>\rank</code> (german spelling)			
<code>\Rg</code>	alias of <code>\rank</code> (german spelling)			
<code>\rowrank</code>	row-rank of operator	<code>rank</code>	$\mathrm{rank} T$	$\mathrm{rank}(T)$
<code>\rrk</code>	alias of <code>\rowrank</code>			
<code>\zrk</code>	alias of <code>\rowrank</code> (german spelling)			
<code>\Zrk</code>	alias of <code>\rowrank</code> (german spelling)			
<code>\colrank</code>	column-rank of operator	<code>rank</code>	$\mathrm{rank} T$	$\mathrm{rank}(T)$
<code>\crk</code>	alias of <code>\colrank</code>			
<code>\srg</code>	alias of <code>\colrank</code> (german spelling)			
<code>\Srg</code>	alias of <code>\colrank</code> (german spelling)			

Table 31: *math* operators for linear algebra

3.14 Number theory

macro	expands to	result
<code>\LeastCommonMultiple</code>	<code>lcm</code>	lcm
<code>\GreatestCommonDivisor</code>	<code>gcd</code>	gcd

Table 32: *prettytex* macros for number theory

macro	arguments	description	example/output
<code>\stirlingone</code>	expression,expression	Stirling number of the first kind	$\left[\begin{smallmatrix} n \\ m \end{smallmatrix} \right]$
<code>\stirlingtwo</code>	expression,expression	Stirling number of the second kind	$\left\{ \begin{smallmatrix} n \\ m \end{smallmatrix} \right\}$
<code>\cycle</code>	array	cycle notation for permutations	$(1 \ 2 \ 3)$

Table 33: *math* macros for number theory

operator	description	example/output		
		alone	w/o parenthesis	w parenthesis
<code>\lcm</code>	least common multiple	<code>lcm</code>		<code>lcm(m, n)</code>
<code>\kgv</code>	alias of <code>\lcm</code> (german spelling)			
<code>\kgV</code>	alias of <code>\lcm</code> (german spelling)			
<code>\gcd</code>	greatest common divisor	<code>gcd</code>		<code>gcd(m, n)</code>
<code>\ggt</code>	alias of <code>\gcd</code> (german spelling)			
<code>\ggT</code>	alias of <code>\gcd</code> (german spelling)			

Table 34: *math* operators for number theory

3.15 Exotic trigonometric functions and relatives

The *Area* functions are the inverses of hyperbolic functions, while the arcus functions are the inverses of the corresponding trigonometric functions.

operator	description	example/output		
		alone	w/o parenthesis	w parenthesis
<code>\Arsinh</code>	Area sinus hyperbolicus	<code>Arsinh</code>	<code>Arsinh x</code>	<code>Arsinh(x)</code>
<code>\Arcosh</code>	Area cosinus hyperbolicus	<code>Arcosh</code>	<code>Arcosh x</code>	<code>Arcosh(x)</code>
<code>\Artanh</code>	Area tangens hyperbolicus	<code>Artanh</code>	<code>Artanh x</code>	<code>Artanh(x)</code>
<code>\Arcoth</code>	Area cotangens hyperbolicus	<code>Arcoth</code>	<code>Arcoth x</code>	<code>Arcoth(x)</code>
<code>\Arsech</code>	Area secans hyperbolicus	<code>Arsech</code>	<code>Arsech x</code>	<code>Arsech(x)</code>
<code>\Arcsch</code>	Area cosecans hyperbolicus	<code>Arcsch</code>	<code>Arcsch x</code>	<code>Arcsch(x)</code>
<code>\sinc</code>	Sinus cardinalis	<code>sinc</code>	<code>sinc x</code>	<code>sinc(x)</code>
<code>\sinnc</code>	Sinus cardinalis (normalised)	<code>si</code>	<code>si x</code>	<code>si(x)</code>
<code>\Si</code>	integral sine	<code>Si</code>	<code>Si x</code>	<code>Si(x)</code>
<code>\Li</code>	integral logarithm	<code>Li</code>	<code>Li x</code>	<code>Li(x)</code>
<code>\arccot</code>	Arcus cotangens	<code>arccot</code>	<code>arccot x</code>	<code>arccot(x)</code>
<code>\arcsec</code>	Arcus secans	<code>arcsec</code>	<code>arcsec x</code>	<code>arcsec(x)</code>
<code>\arccsc</code>	Arcus cosecans	<code>arccsc</code>	<code>arccsc x</code>	<code>arccsc(x)</code>
<code>\sech</code>	Secans hyperbolicus	<code>sech</code>	<code>sech x</code>	<code>sech(x)</code>
<code>\csch</code>	Cosecans hyperbolicus	<code>csch</code>	<code>csch x</code>	<code>csch(x)</code>
<code>\atan</code>	atan2-map	<code>atan2</code>		<code>atan2(x, y)</code>
<code>\ld</code>	Logarithmus dualus (base-2 log)	<code>ld</code>	<code>ld x</code>	<code>ld(x)</code>

Table 35: *math* operators for exotic trigonometric functions

3.16 Overset relations

Sometimes, it is useful to reference a result/equations etc in a relation by setting it over the relational symbol. These macros provide shortcuts to do precisely that. You can either define your own or use the list of common symbols below. We provide both reference to theorems et cetera as well as equations (these macros have the same name as the regular reference ones but with the `eq` prefix).

macro	expandsto	result
<code>AlmostEverywhere</code>	<code>ae</code>	<code>ae</code>

Table 36: *prettytex* macros for overset relations

macro	arguments	description
<code>\hateq</code>		equals-sign with hat ($\hat{=}$), used for congruence or similar
<code>\refrel</code>	reference, expression	relation with reference overset to expression
<code>\refqual</code>	reference	reference overset of $=$
<code>\refleq</code>	reference	reference overset of $\leq$
<code>\refle</code>	reference	reference overset of $<$
<code>\refgeq</code>	reference	reference overset of $\geq$

<code>\refge</code>	reference	reference overset of $>$
<code>\refimpl</code>	reference	reference overset of $\impl$
<code>\refimpliedb</code>	reference	reference overset of $\Leftarrow$
<code>\refeqv</code>	reference	reference overset of $\iff$
<code>\refiso</code>	reference	reference overset of $\cong$
<code>\eqrefrel</code>	reference, expression	relation with equation reference overset to expression
<code>\eqrefqual</code>	reference	equation reference overset of $=$
<code>\eqrefleq</code>	reference	equation reference overset of $\leq$
<code>\eqrefle</code>	reference	equation reference overset of $<$
<code>\eqrefgeq</code>	reference	equation reference overset of $\geq$
<code>\eqrefge</code>	reference	equation reference overset of $>$
<code>\eqrefimpl</code>	reference	equation reference overset of $\impl$
<code>\eqrefimpliedb</code>	reference	equation reference overset of $\Leftarrow$
<code>\eqrefeqv</code>	reference	equation reference overset of $\iff$
<code>\eqrefiso</code>	reference	equation reference overset of $\cong$
<code>\aerel</code>	expression	relation with AlmostEverywhere overset
<code>\aeq</code>		almost everywhere equal
<code>\aequal</code>		alias of <code>\aeq</code>
<code>\aeleq</code>		almost everywhere less than or equal
<code>\aele</code>		almost everywhere less than
<code>\aegeq</code>		almost everywhere greater than or equal
<code>\aege</code>		almost everywhere greater than